The operation consists of first taking the logarithm of the function $f(x)$ (to base e) then differentiating is called **logarithmic differentiation** and its result

$$\boxed{\frac{d}{dx}(\log f(x)) = \frac{f'(x)}{f(x)}}.$$

is called the logarithmic derivative of $f(x)$.

The advantage in this method is that the calculation of derivatives of complicated functions involving products, quotients or powers can often be simplified by taking logarithms.

Example 10.21

Find the derivative of $y = \sqrt{x^2+4}\,.\sin^2 x\,.2^x$

Solution

Taking logarithm on both sides and using the law of logarithm,

we have, $\log y = \frac{1}{2}\log(x^2+4) + 2\log(\sin x) + x\log(2).$

$$\text{This implies, } \frac{y'}{y} = \frac{1}{2}\frac{2x}{x^2+4} + 2.\frac{\cos x}{\sin x} + \log 2$$

$$= \frac{x}{x^2+4} + 2\cot x + \log 2$$

$$\text{Therefore, } y' = \frac{dy}{dx} = y\left(\frac{x}{x^2+4} + 2\cot x + \log 2\right).$$

Example 10.22

Differentiate : $y = \dfrac{x^{\frac{3}{4}}\sqrt{x^2+1}}{(3x+2)^5}$.

Solution

Taking logarithm on both sides of the equation and using the rules of logarithm we have,

$\log y = \frac{3}{4}\log x + \frac{1}{2}\log(x^2+1) - 5\log(3x+2).$

Differentiating implicitly

$$\frac{y'}{y} = \frac{3}{4x} + \frac{1}{2}\frac{2x}{(x^2+1)} - \frac{5\times 3}{3x+2}$$

$$= \frac{3}{4x} + \frac{x}{(x^2+1)} - \frac{15}{3x+2}$$

$$\text{Therefore, } \frac{dy}{dx} = y' = \frac{x^{\frac{3}{4}}\sqrt{x^2+1}}{(3x+2)^5}\left[\frac{3}{4x} + \frac{x}{x^2+1} - \frac{15}{3x+2}\right].$$

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